

TRACE Addendum: Agent Mandates as a Formal Sub-Token Schema

Proposed addition to Layer 1 (Identity) Drafted against Position Paper v0.3 (June 2026).

1. The gap this closes

v0.3 names the mechanism without specifying its content: "Recursive Sub-Tokens enable high-velocity delegation without losing chain of custody." A sub-token is, in practice, a mandate — a bounded grant of authority from one party to an agent. But nothing in the current paper defines what a sub-token actually carries, which means two TRACE-compliant implementations could both claim conformance while producing sub-tokens that answer different questions, or none.

This addendum proposes that every sub-token be required to answer three questions, explicitly and machine-checkably:

- **Who authorised this** — the issuing identity, itself a TRACE-registered DID bound to a verified legal entity or a human acting within a verified role at one.
- **To do what** — a scope, specific enough that a resource checking the token can determine whether a given requested action falls inside or outside it.
- **Until when** — an expiry, plus a live revocation status independent of that expiry.

2. Proposed schema: sub-token as Verifiable Credential

Rather than defining a new credential format, sub-tokens should be issued as W3C Verifiable Credentials, with TRACE defining a specific credential type (**TraceAgentMandate**) and a required claim set:

Field	Required	Description
issuer	Yes	DID of the granting identity (must itself be TRACE-registered and in good standing at time of issuance)
subject	Yes	DID of the agent instance receiving the mandate

Field	Required	Description
<code>scope</code>	Yes	A structured claim describing the bounded action set — see §3
<code>validFrom</code> / <code>validUntil</code>	Yes	Standard VC temporal bounds
<code>revocationRegistry</code>	Yes	Pointer to where this specific mandate's live status can be checked — see §4
<code>delegable</code>	No, default false	Whether the subject may issue further sub-tokens narrowing this mandate's scope to another agent
<code>collateralRef</code>	No	Pointer to any collateral posted against this specific mandate, where risk tier requires it — see §5

Building on an existing standard rather than a bespoke format means a mandate is independently verifiable by any party that already speaks Verifiable Credentials, without requiring TRACE-specific tooling to parse the core claim.

3. Scope needs a controlled structure, not free text

A scope field that's just a string ("can draft invoices") is unenforceable at machine speed — the whole point of the exercise is that a resource can evaluate, in real time, whether a requested action is inside or outside the grant. This addendum doesn't attempt to define a universal action taxonomy here — that's a larger, separate piece of work, plausibly the next addendum — but flags the requirement: scope should resolve to a structured, enumerable set (action type, resource class, bound values where relevant, e.g. a currency ceiling), not prose.

4. Revocation authority must be split, and isn't currently

v0.3's Layer 3 governance already gets the two-speed principle right at the identity level: automated Circuit Breakers for real-time containment, human supermajority for permanent revocation. Both, as written, act on an agent's overall registry status.

This addendum proposes a third, distinct revocation authority: **the mandate issuer**, acting unilaterally on their own sub-token, without invoking TRACE governance at all. If Party A issues a sub-token to Agent X, Party A must be able to revoke that specific sub-token — and only that one — without the Foundation's involvement, without a Circuit Breaker firing, and without affecting any other mandate Agent X holds from any other issuer. This is not a governance action; it is the ordinary exercise of authority the issuer already had by virtue of being the issuer.

Concretely, this means the **revocationRegistry** field in §2 cannot simply point at TRACE's central identity-status registry. It needs to point at an issuer-controlled revocation list (a standard VC status-list mechanism is sufficient) that the issuer alone can write to. Three revocation layers now exist, cleanly separated by who can pull each lever:

Layer	Actor	Scope of effect	Speed
Mandate revocation	The issuer	One sub-token	Immediate, issuer-initiated
Circuit Breaker	Automated / Foundation	Agent's registry standing	Real-time, automated trigger
Governance revocation	Human supermajority	Agent's registry standing, permanent	Deliberate, slow

5. Trust tier and collateral should attach to the mandate, not only the agent

v0.3's Layer 2 gates access by the agent's tier; Layer 4 sizes collateral by whether the agent is classified high-risk. Both are currently agent-level properties. This addendum proposes that tier and collateral be computable per active mandate: an agent's effective trust tier for a given action is a function of both its baseline identity tier and the scope of the specific mandate it's acting under. A narrow, low-value mandate should not require the same collateral posture as a broad one, even from the same agent identity. Where **collateralRef** (§2) is populated, the collateral required should be sized to the mandate's declared scope, not fetched from a single agent-level figure.

This has a practical consequence for Layer 2's real-time lookup: a resource checking an agent must check not "is this agent registered and in good standing" but "does this agent hold a currently valid, unrevoked mandate whose scope covers this specific requested action, and is any required collateral for that mandate posted." That's a heavier query than the current paper implies, and worth stating plainly rather than discovering at implementation time.

6. What this addendum does not solve

Formalizing the sub-token schema makes a mandate authentic, traceable, and revocable. It does not establish that the human issuing a mandate on behalf of a legal entity exercised genuine judgment in doing so, as opposed to approving a request they did not meaningfully evaluate. TRACE's Identity layer anchors trust in the issuer being a verified legal entity; it says nothing about the quality of the decision that entity's representative made before signing. That is a governance and process question for the issuing organisation, not something a registry can attest to, and this addendum makes no claim otherwise. It is scoped here deliberately, to avoid the schema quietly implying a guarantee it cannot make.

7. Open questions for the next pass

- Does **delegable** need depth limits (a mandate that permits re-delegation indefinitely down a chain is a different risk profile than one capped at one hop)?
- How does a resource verifying a mandate handle an issuer whose own TRACE registration is revoked *after* the mandate was issued but *before* it expires — does the mandate die with the issuer's standing, or only if the issuer explicitly revokes it?
- Does the standard VC status-list mechanism proposed in §4 need TRACE-specific extensions, or is it sufficient as-is?

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